

A NAANMUDHALVAN PROJECT REPORT

SB8067 – SALESFORCE DEVELOPER

GARAGE MANAGEMENT SYSTEM

Submitted by

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ARASAMPATTI – 622 507

BONAFIDE CERTIFICATE

Certified that this project report “**GARAGE MANAGEMENT SYSTEM IN SALESFORCE**” is the bonafide work of “**P.SAKTHI POORNIMA (912422104035), M.RAJESHWARI (912422104031), B.SRI RATHNALAKSHMI (912422104041), S.SWATHI (912422104048)**” who carried out the mini project Work under my supervision.

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ACKNOWLEDGEMENT

With warm hearts and immense pleasure, we thank the almighty for his grace and blessing bestowed on us, which drove us to the successful completion of this project. We take this opportunity to express our sincere thanks to the significant contribution given by the management of our College Chairman “**KALVI VALLAL Mrs.Pichappa Valliammai**”, who is guiding light for all activities in our college.

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We also convey our humble Finally, we thank our family members who have spent their wealth for our studies and have motivated us to complete this project successfully.

S.NO	TITLE	PG.NO
1.	Abstract	5
2.	Introduction	5
3.	Objectives	6
4.	System Requirements	6
5.	Technologies Used	6
6.	System Design	7
7.	Workflow Description	7
8.	Workflow Diagram	8
9.	Implementation Steps	8
10.	Expected Outcomes	19
11.	Advantages	20
12.	Future Enhancements	20
13.	Conclusion	20
14.	References	20

1. Abstract

The Garage Management System (GMS) developed on Salesforce CRM aims to streamline and automate the operations of automobile service centers and garages. The system helps manage customer details, vehicle records, service requests, job cards, inventory tracking, and billing, all within Salesforce's cloud-based environment. By integrating automation tools, workflows, and reports, the system ensures efficient service handling, reduces manual effort, and enhances customer satisfaction. The solution leverages Salesforce's standard objects, custom apps, Lightning components, and automation flows to create a centralized and smart garage management platform.

The Garage Management System built on the Salesforce platform is designed to streamline and automate daily garage operations such as customer service, vehicle service tracking, inventory management, and billing. Leveraging Salesforce's powerful CRM capabilities, the system enables efficient handling of customer profiles, service history, and appointment scheduling while ensuring real-time tracking of ongoing repairs and spare parts availability.

Service advisors can easily generate service estimates and invoices, assign tasks to technicians, and monitor job progress using customizable dashboards and workflows. The solution enhances transparency, reduces manual errors, and improves customer satisfaction through automated notifications and service updates. With Salesforce's cloud-based architecture, the system ensures high scalability, data security, and integration flexibility with other business applications, making it a comprehensive and modern approach to managing garage operations.

2. Introduction

A Garage Management System is essential for modern automotive service centers to efficiently manage their day-to-day operations. Traditional garages often rely on manual processes for handling customer appointments, vehicle service details, inventory tracking, and billing, which can lead to errors, delays, and poor customer experience. To overcome these limitations, the Garage Management System is implemented using **Salesforce**, a leading cloud-based Customer Relationship Management (CRM) platform.

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To overcome these limitations, the Garage Management System is implemented using **Salesforce**, a leading cloud-based Customer Relationship Management (CRM) platform. The Salesforce-based solution not only enhances operational productivity but also improves customer satisfaction through **real-time service tracking, automated notifications, and secure data management**. It empowers the garage staff to deliver faster, more organized, and professional service while enabling managers to make data-driven decisions.

3. Objective

- To **automate the complete workflow** of garage operations including customer management, service booking, job assignment, and billing.
- Maintain **accurate customer and vehicle service history** for future reference and personalized service.
- To enable **real-time tracking of service status** for both customers and garage staff.
- To **manage spare parts inventory efficiently** and prevent stock shortages or delays.
- To **reduce manual errors and paperwork** by digitizing all service-related activities.
- To improve **customer satisfaction** through automated updates, reminders, and faster service delivery.
- To provide **dashboard and reports** for managers to analyze performance, revenue, and service trends.
- To ensure **secure storage and easy access to data** using Salesforce's cloud-based platform.
- To support **scalability and easy integration** with other business tools like SMS, Email, or Payment gateways.

4. System Requirements

A. Software Requirements

Salesforce Developer Edition
Google Chrome / Microsoft Edge
Windows / macOS
Internet Connectivity
Trailhead Playground

B. Hardware Requirements

Processor: Intel i3 or higher
RAM: Minimum 4 GB
Storage: Minimum 10 GB
Stable Internet connection
POS Device / Card Swipe

5. Technology Used

1. **Salesforce Lightning:** Core platform for building the Garage Management System
2. **Custom Objects (Customer, Appointment, Service Record, Billing, Feedback):** Define database structure
3. **Workflow Rules / Salesforce Flow:** Automate tasks like updating records or sending notifications
4. **Lightning App Builder, Page Layouts:** To create user-friendly interfaces

5. **Report and Dashboards:** For service performance and payment analysis
6. **Lookup & Master-Detail Fields :**To link records between objects
7. **Profiles, Sharing Settings, Role Hierarchy:** To manage access and permissions
8. **Cloud-based (SAAS):** Hosted and run entirely on Salesforce cloud
9. **Lightning Record Pages, Data Import Wizard:** For creating and managing data records
10. **Salesforce/SetupObject Manager:** Environment used for configuration and customization.

6. System Design

The system follows a client–server architecture using Salesforce’s multi-tenant cloud model:

1. **User Interface Layer:** Built using Lightning Pages and Components.
2. **Application Logic Layer:** Managed using Apex Classes, Flows, and Validation Rules.
3. **Database Layer:** Implemented using Salesforce Custom Objects (Customer, Vehicle, Service Request, Job Card, etc.).

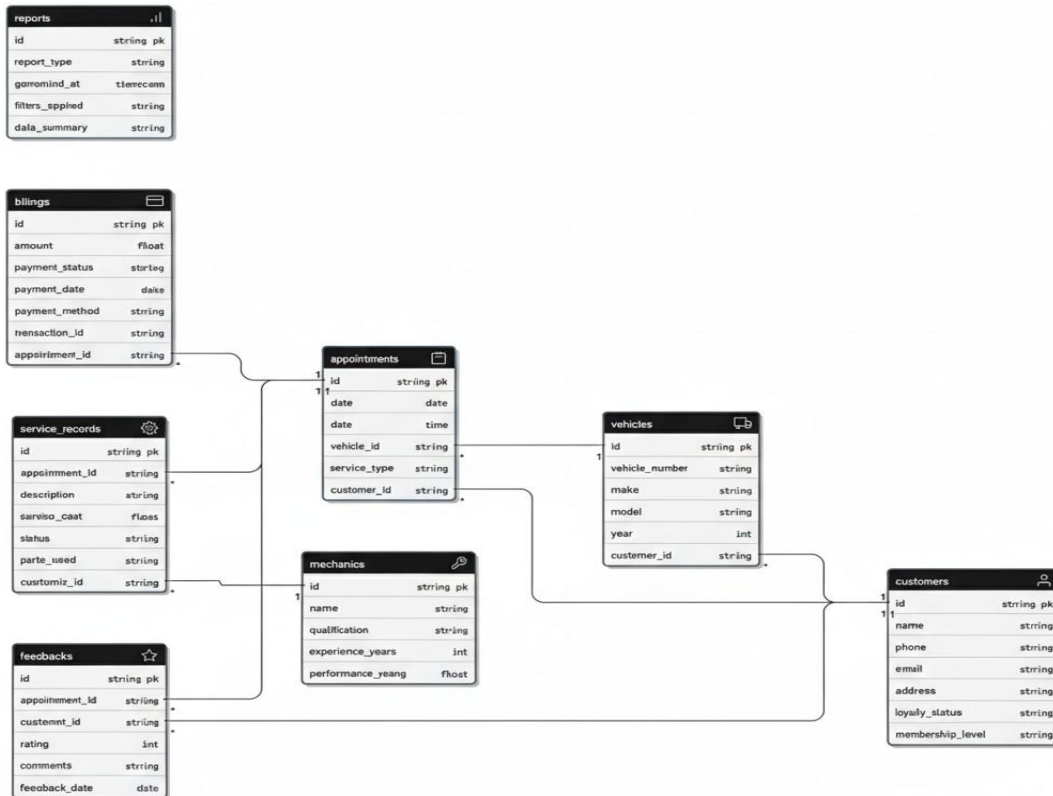
Entity Relationship Diagram (ERD):

- Customer → Vehicle (One-to-Many)
- Vehicle → Service Request (One-to-Many)
- Service Request → Job Card (One-to-One)
- Job Card → Mechanic (Many-to-One)
- Service Request → Billing (One-to-One)

7. Workflow Description

- **Customer Registration:** A new customer is added via Salesforce using the “Customer” object.
- **Vehicle Addition:** The customer’s vehicle details are entered and linked to the customer record.
- **Service Request Creation:** A service request is logged with complaint details and assigned to a mechanic.
- **Job Card Generation:** When the service starts, a job card is generated showing tasks and estimated cost.
- **Inventory Update:** Parts used during service are deducted from the inventory automatically.
- **Billing & Invoice:** Once service is completed, the system generates a bill with service and parts cost.
- **Reports Generation:** Management can view dashboards showing daily income, number of services, etc.

8.Workflow Diagrams



9. Implementation Steps

1. SALESFORCE ACCOUNT CREATION
2. OBJECT CREATION
3. CREATING CUSTOM OBJECT TABS
4. CREATING A LIGHTNING APP
5. CREATION OF FIELDS IN OBJECTS
6. CREATING VALIDATION RULES
7. CREATING DUPLICATE RULES
8. CREATING A PROFILES
9. CREATING ROLES
10. CREATING USERS
11. CREATING PUBLIC GROUP
12. SETTING UP SHARING SETTINGS
13. CREATING FLOW

14. DEVELOP APEX TRIGGERS

15. CREATING A REPORT

16. CREATING A DASHBOARD

With Screenshots:

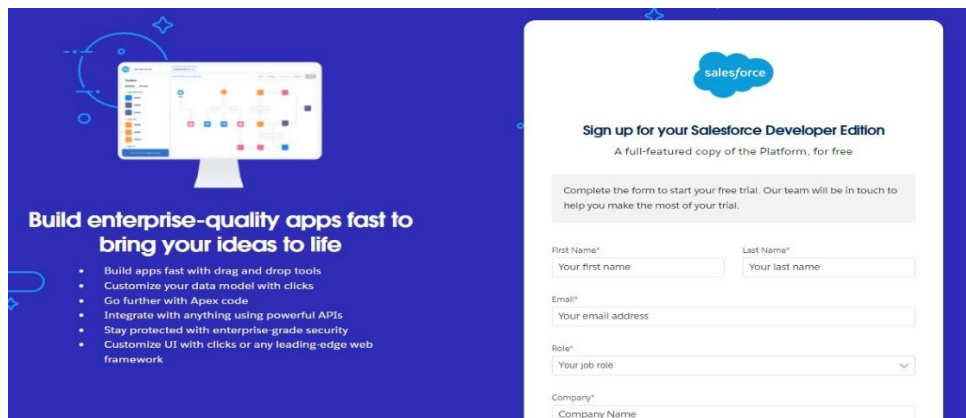


Fig 1.1 Sign up for salesforce developer

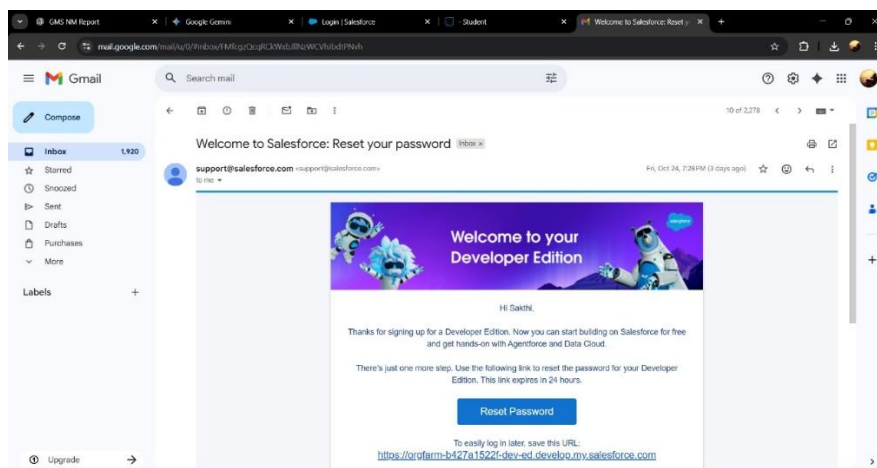


Fig 1.2 Activation of salesforce account

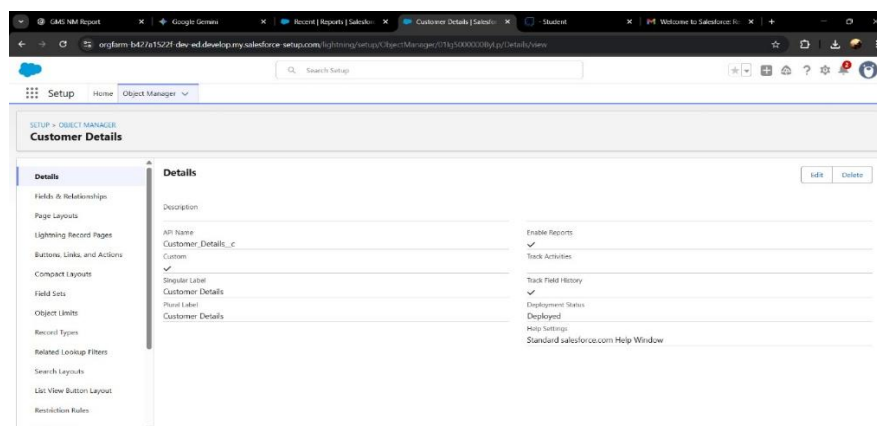


Fig 2.1 Customer object

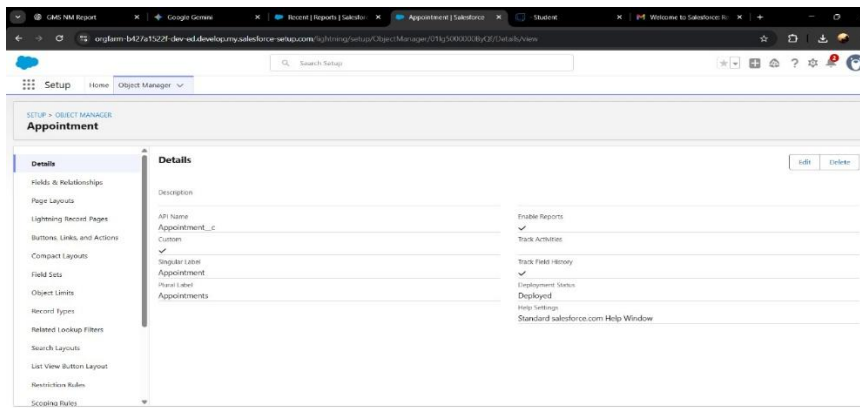


Fig 2.2 Appointment object

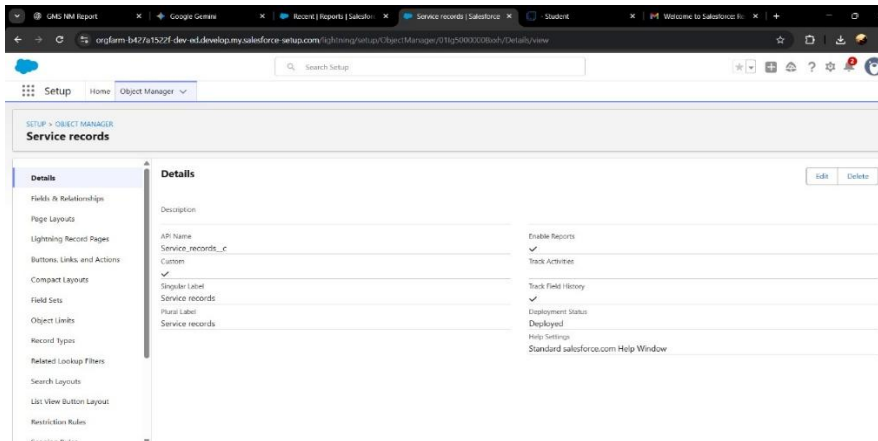


Fig 2.3 Service record object

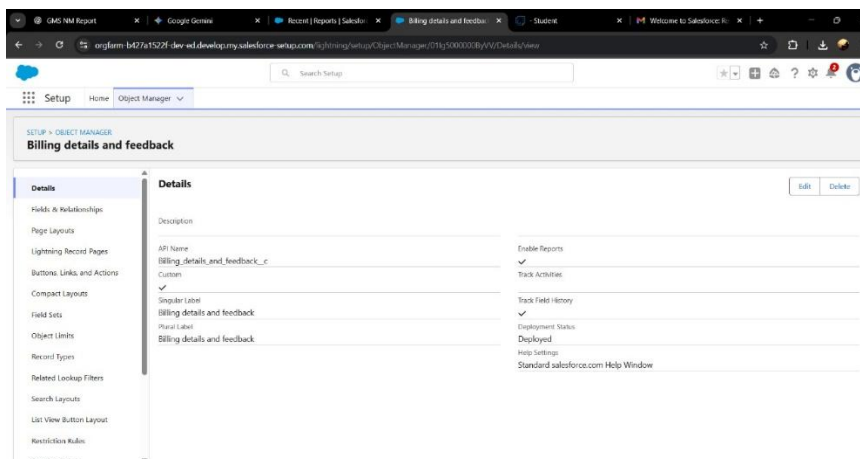


Fig 2.4 Billing details and feedback object

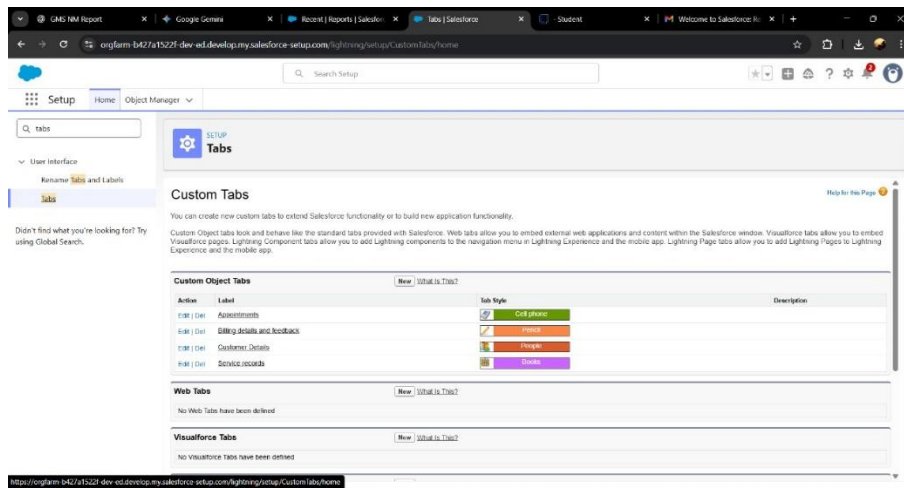


Fig 3.1 Creating custom tabs for appointments, billing details and feedback, customer details, service records

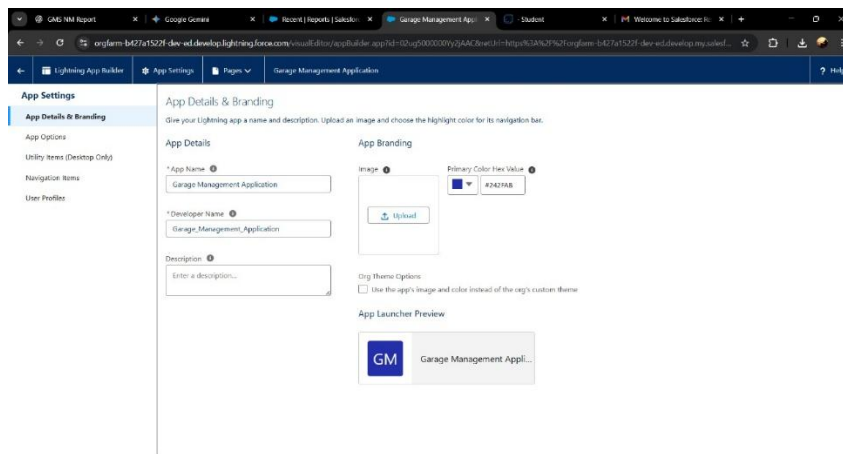


Fig 4.1 Lightning app for garage management application

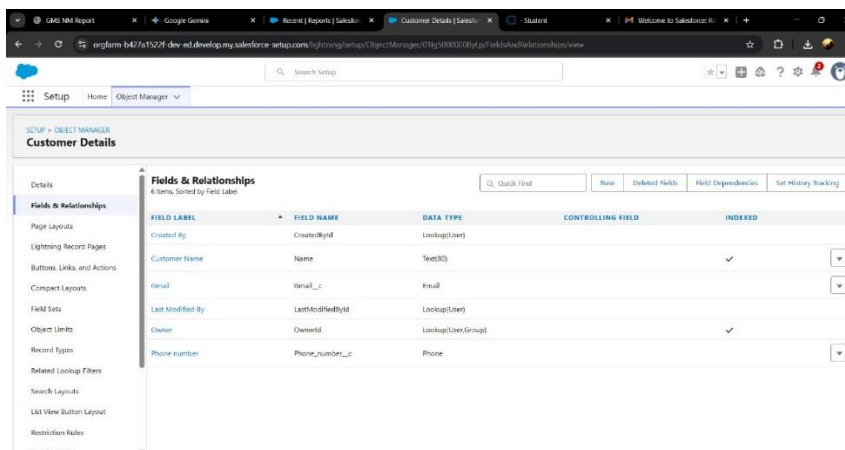


Fig 5.1 Fields of customer details

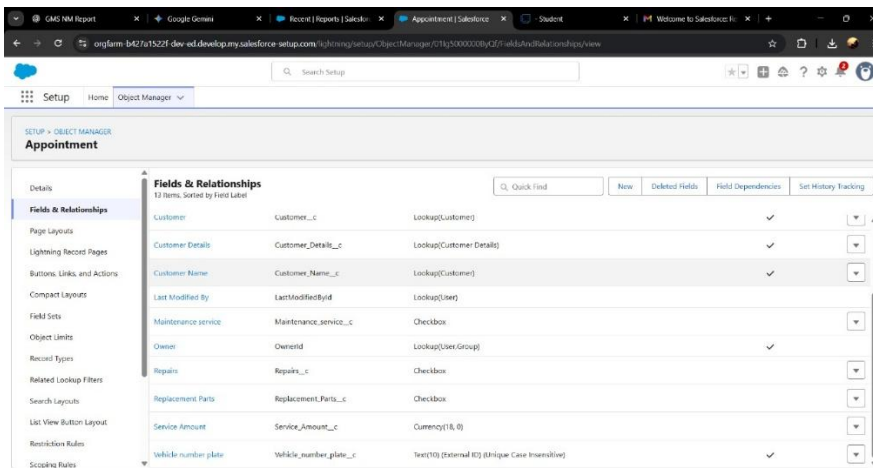


Fig 5.2 Fields of appointment

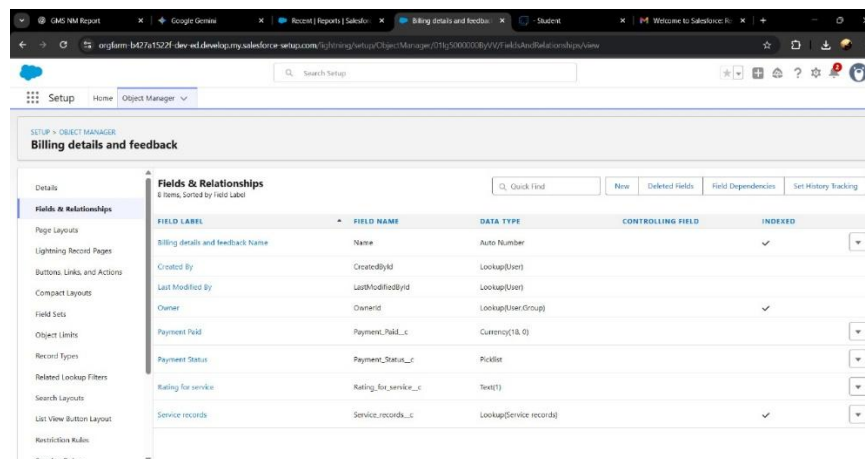


Fig 5.3 Fields of billing details and feedback

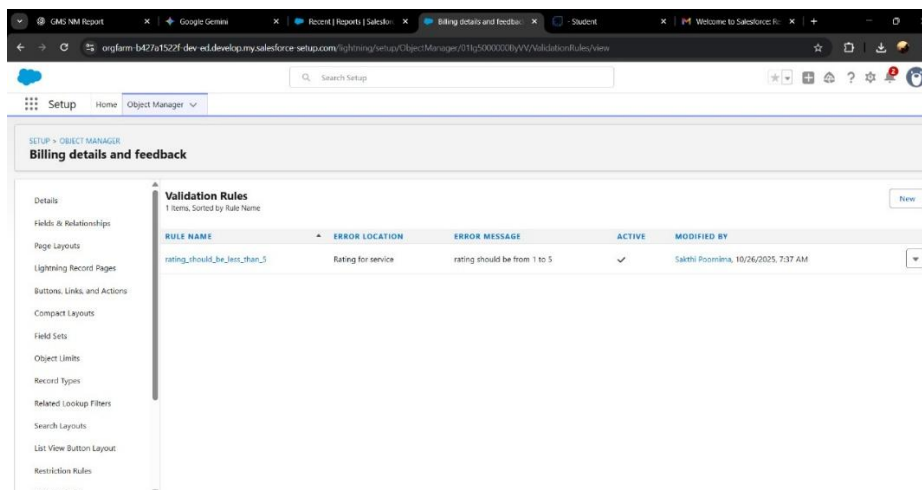


Fig 6.1 Validation rules billing details and feedback

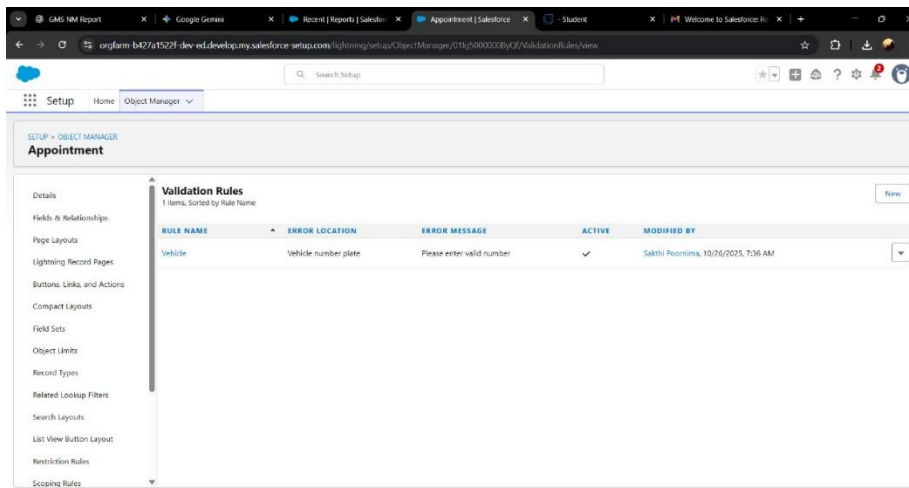


Fig 6.2 Validation rules for appointment object

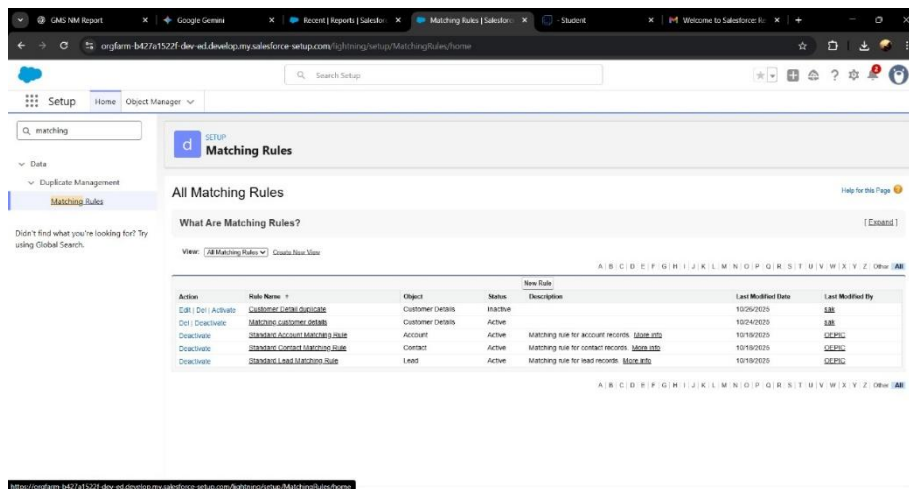


Fig 7.1 Matching rules and duplication rules of customer details

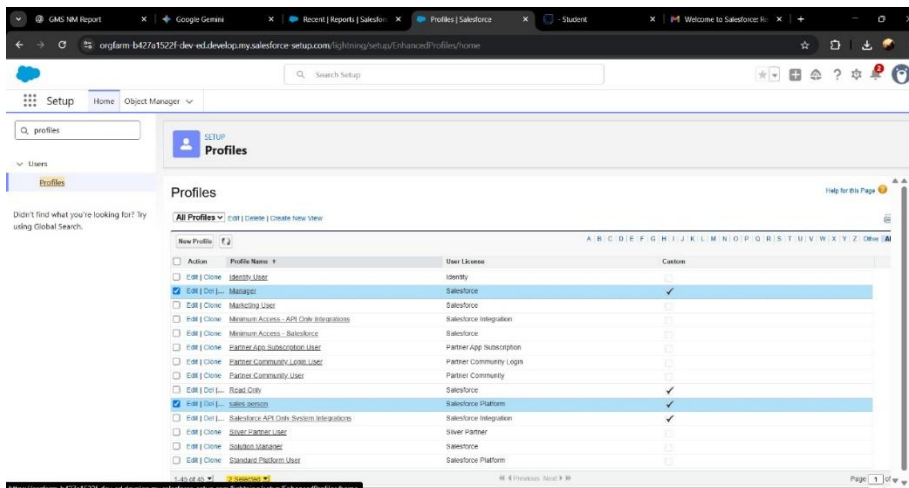


Fig 8.1 Setting up profile for manager and salesperson

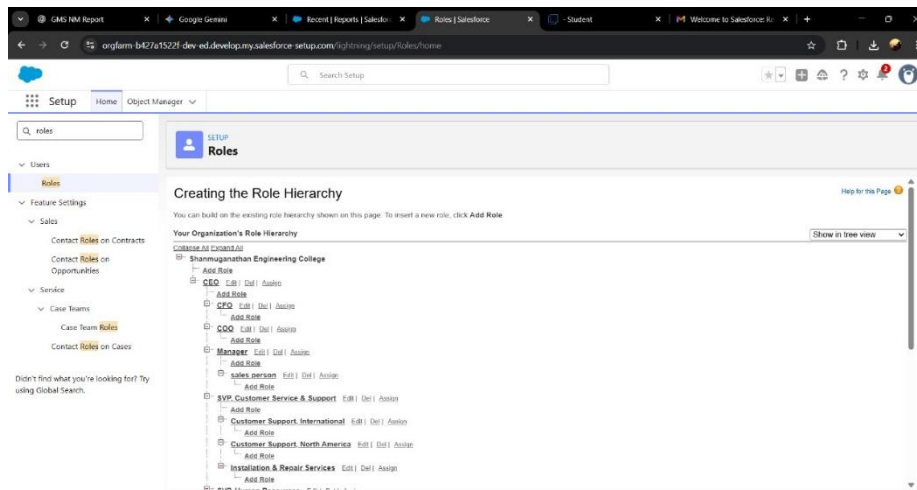


Fig 9.1 Creating rule hierarchy for manager and salesperson

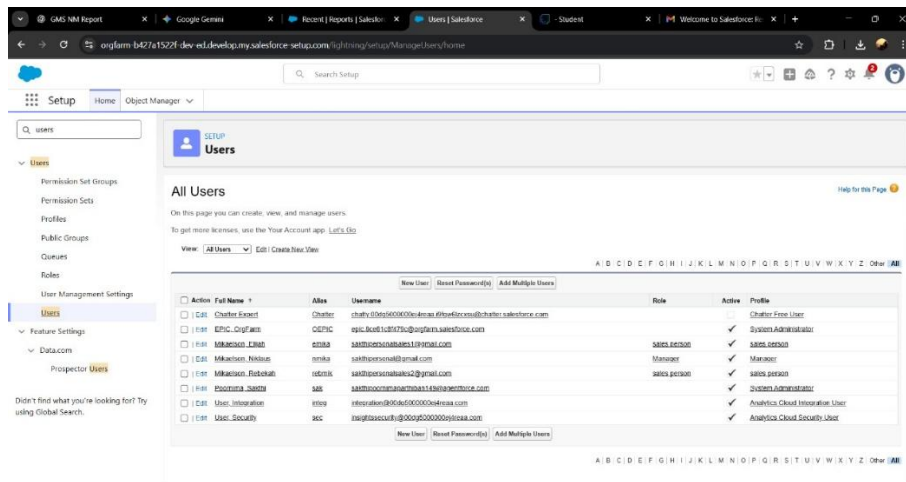


Fig 10.1 Creating users

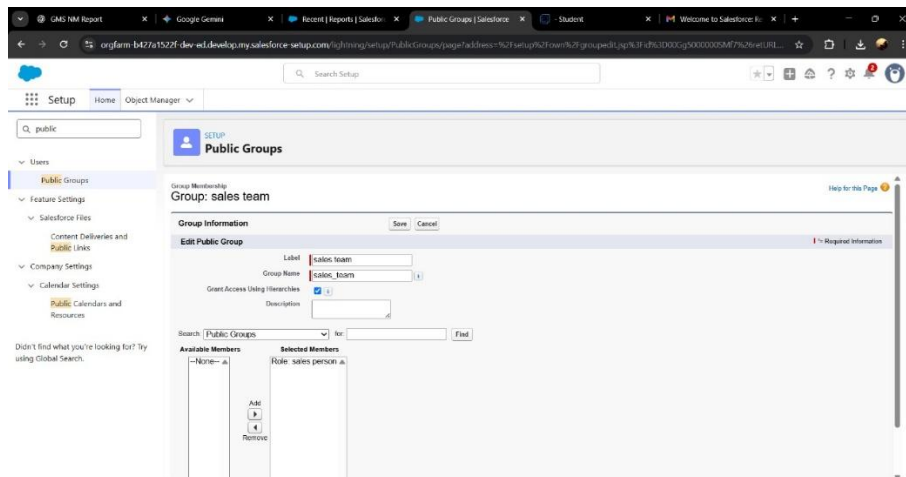


Fig 11.1 Creating public groups

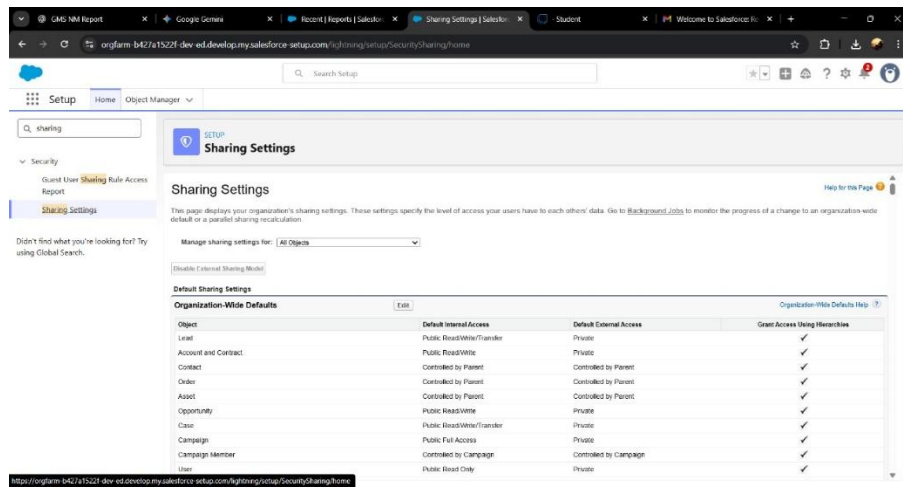


Fig 12.1 Creating sharing setting

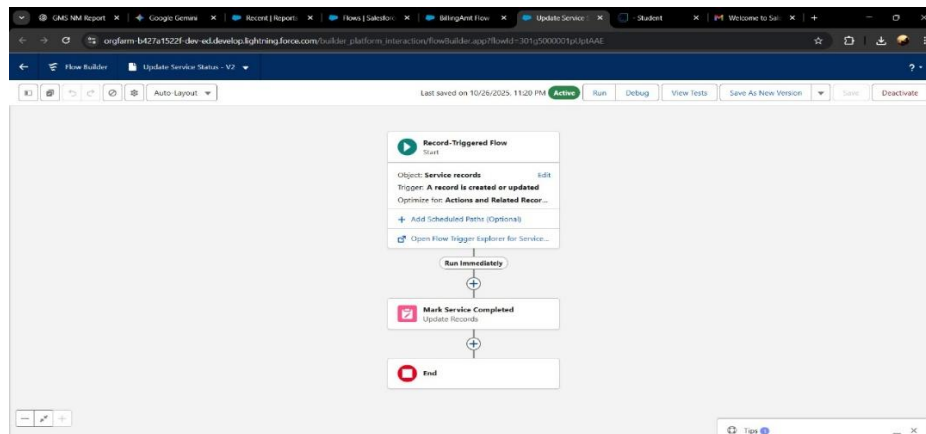


Fig 13.1 Creating flow for update service status

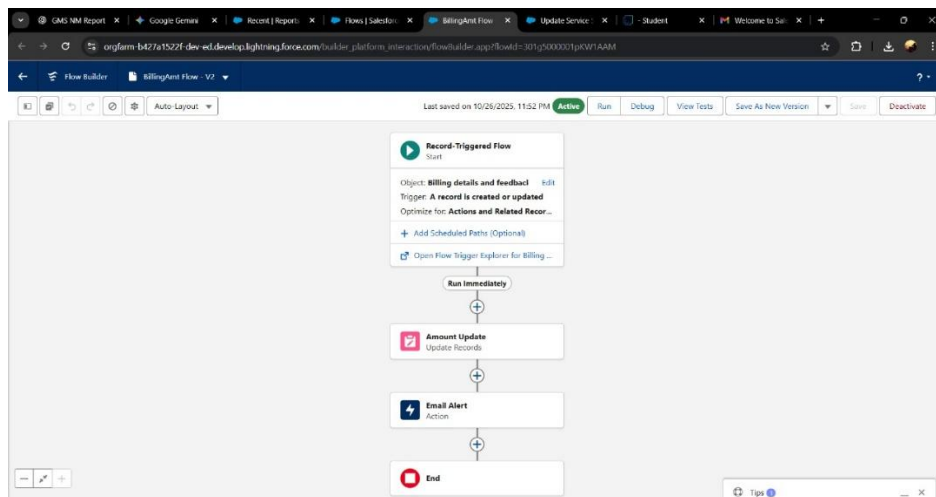


Fig 13.2 Creating flow for billing amount flow

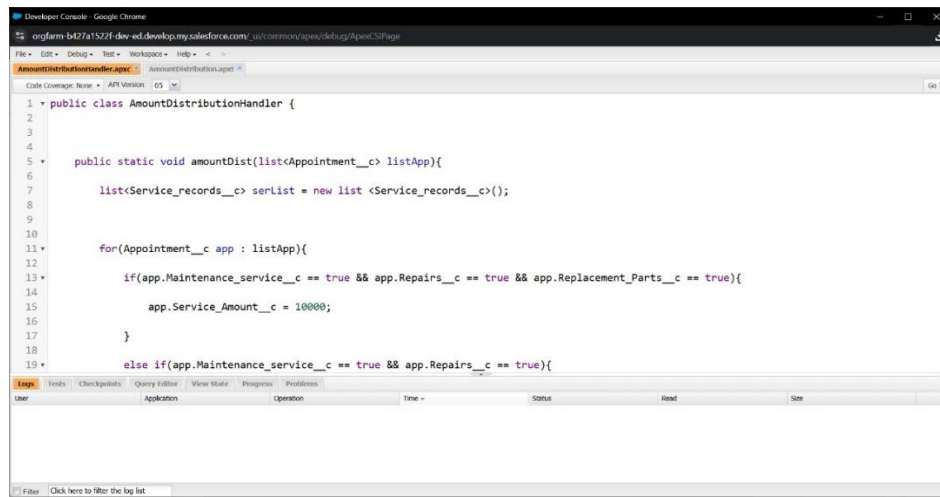


Fig 14.1 Creating apex classes

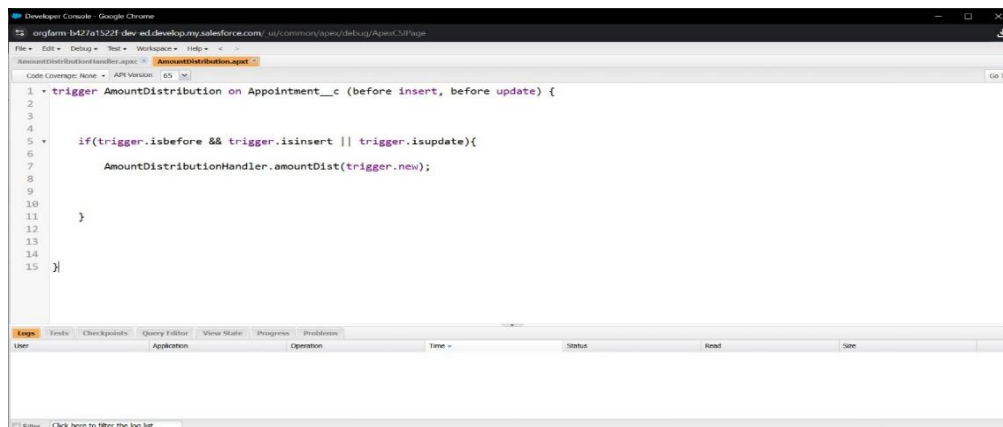


Fig 14.2 Creating apex triggers

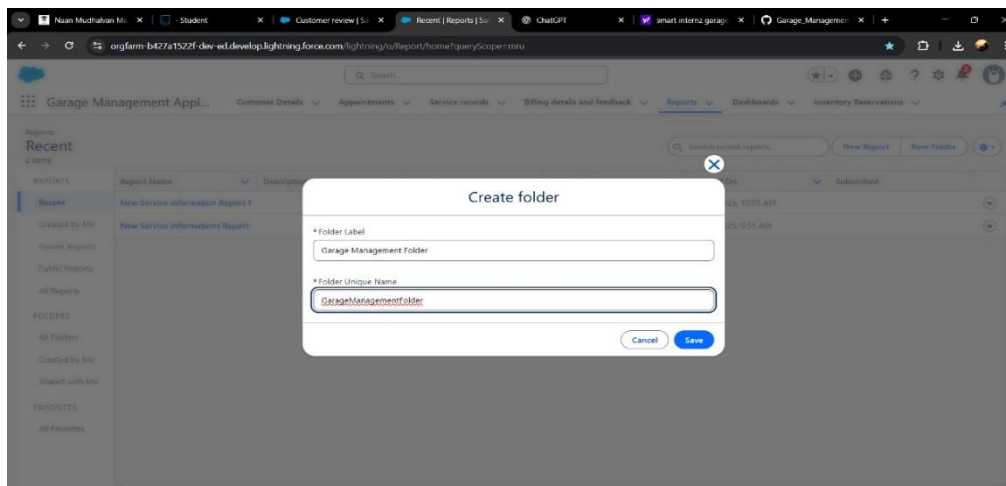


Fig 15.1 Creating report folder

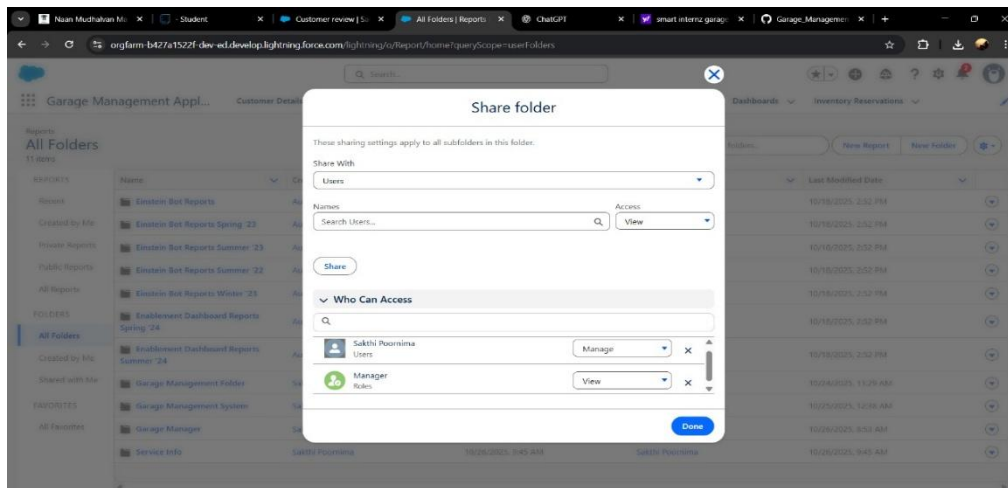


Fig 15.2 Sharing report folder

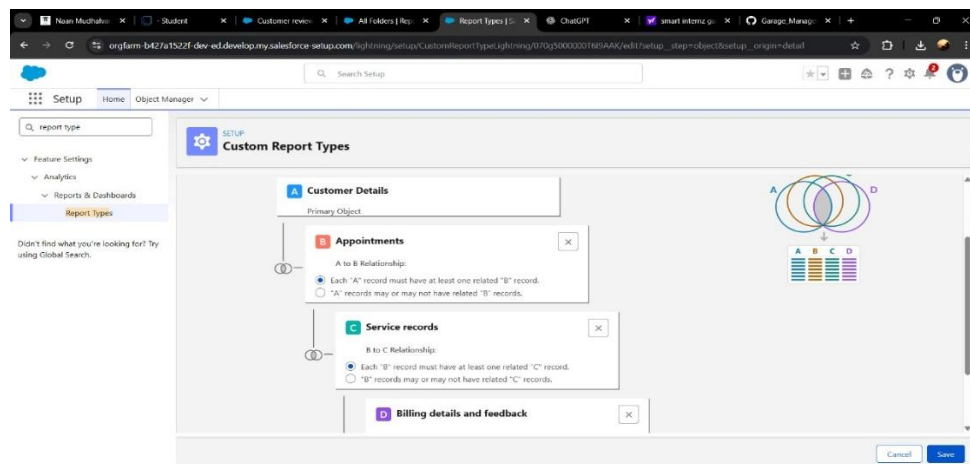


Fig 15.3 Assigning report types

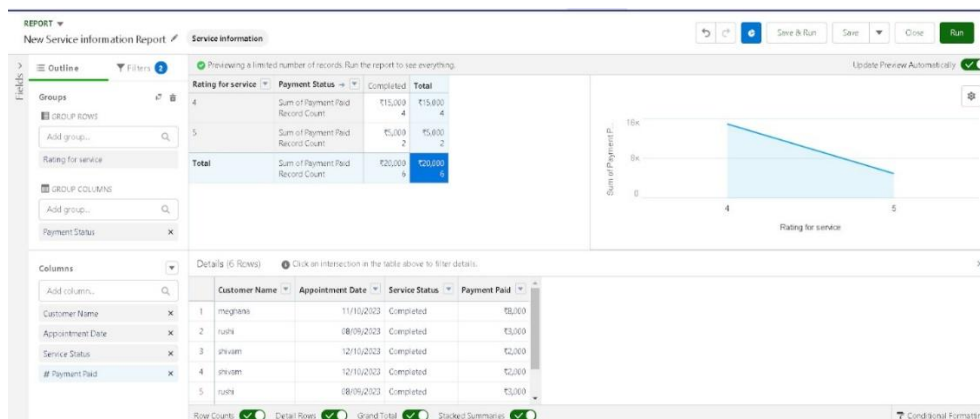


Fig 15.4 Creation of service information report

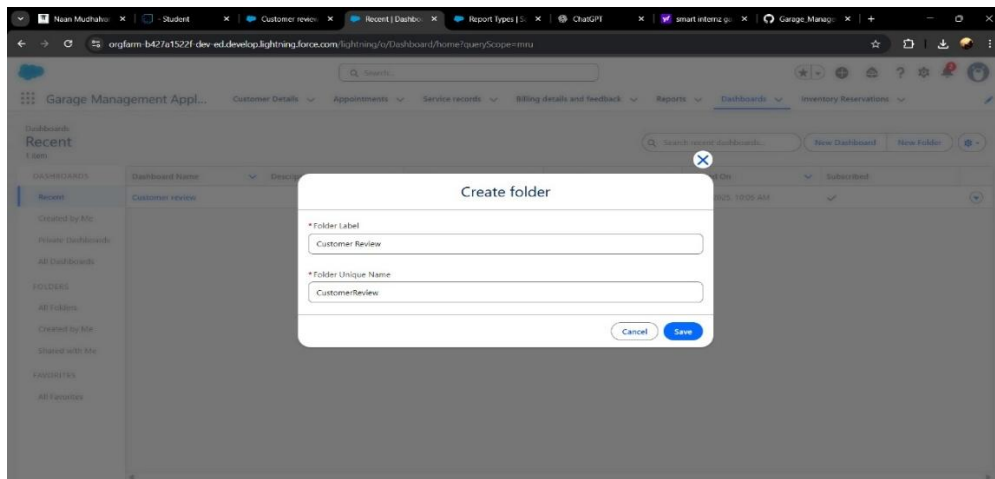


Fig 16.1 Creating dashboard folder

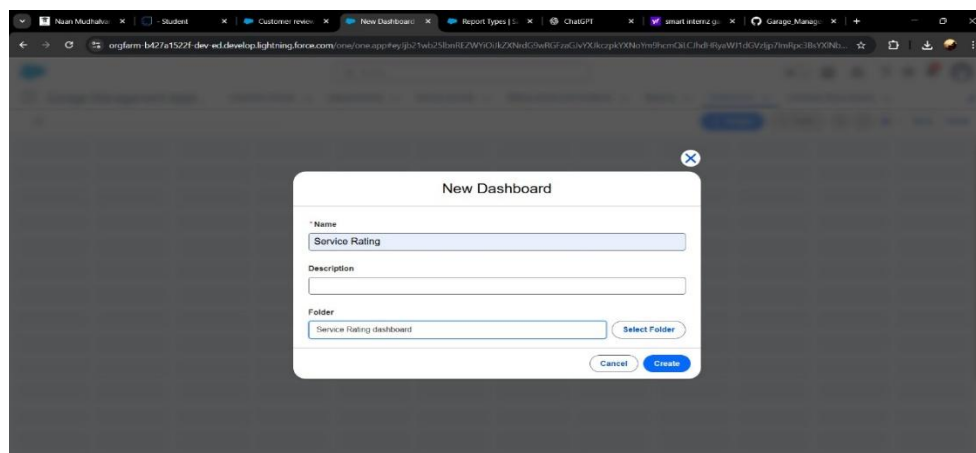


Fig 16.2 Creating service rating dashboard

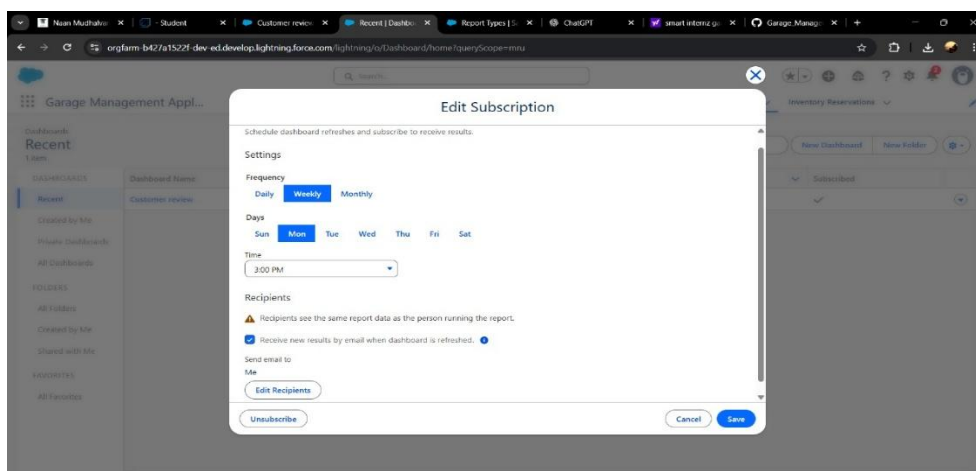


Fig 16.3 Subscription plan

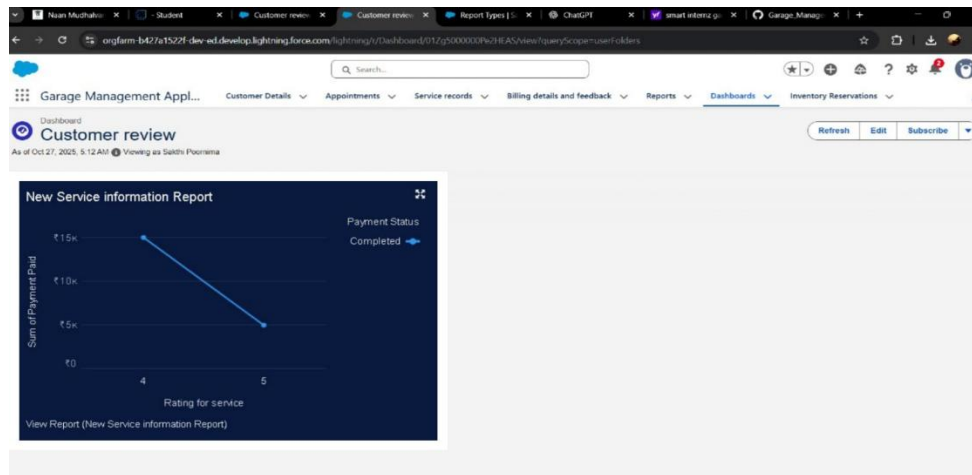


Fig 16.4 Garage management system Dashboard

10. Expected Outcomes

- Fully automated and paperless garage operation.
- Reduced turnaround time for each service request.
- Centralized system for customers, vehicles, and invoices.
- Increased customer satisfaction with transparent billing and updates.
- Real-time insights into business performance through dashboards.

11. Advantages

- Cloud-based, accessible from anywhere.
- Highly secure and customizable using Salesforce CRM features.
- Reduces manual data entry and human error.
- Integrates with Salesforce mobile app for on-the-go access.
- Scalable — can handle multiple garages or branches.

12. Future Enhancements

- Integration with IoT sensors to auto-detect vehicle service requirements.
- SMS or WhatsApp notifications for service updates.
- Online booking portal integrated with Salesforce Experience Cloud.
- AI-based service prediction using Salesforce Einstein.

13. Conclusion

The Garage Management System in Salesforce demonstrates the power of cloud CRM in automating business processes beyond sales. The project successfully digitizes garage operations by managing customers, vehicles, services, and billing efficiently. Using Salesforce's automation tools, the system enhances accuracy, transparency, and customer trust. The project can be further expanded with AI and IoT integrations to create a next-generation smart garage solution.

The Garage Management System on Salesforce provides an efficient, automated, and user-friendly platform to manage all garage operations in one place. By leveraging Salesforce's cloud capabilities, the system ensures secure data storage, real-time access, and seamless workflow automation for activities such as customer registration, appointment scheduling, vehicle service tracking, billing, and inventory management.

This project demonstrates how Salesforce can be customized using Lightning components, custom objects, workflows, and reports to meet the specific needs of automotive service centers. The system reduces manual work, minimizes human error, and enhances customer satisfaction through faster service and transparent communication.

14. References

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- **Lightning Web Components Documentation:**

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